Critical Memory in Price–Score Learners: Exponential Late-Credit Gaps in Routing Markets

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Abstract

A router can make a profitable price cut hard to learn without making it unprofitable. We study a price–score rule that temporarily penalizes a provider after a cut. The resulting MDP has two boundaries: a rational memory threshold above which staying high is optimal and a finite-time threshold above which fresh evidence for a profitable persistent cut is exponentially rare. After downstream values become usable, a learner cutting with conditional probability at most q sees a fresh length-M path within T steps with probability at most Tq^M. A path-equivalent commitment option reduces path proposal time to linear while preserving feasible primitive paths and optimal value. In a frozen calibrated game, primitive Q-learning chooses the rational action in 1/20 seeds at $M = 7$, versus 19/20 with the option; paired regret falls by 0.0643 (95% interval $[0.0493, 0.0755]$). All seeds cover every state–action pair, but failed learners stop revisiting deep cut states; ordered batch Bellman sweeps recover the optimum in 20/20. The option overcuts beyond the rational boundary. The result isolates a temporal credit mechanism, not live-provider collusion.

1 Introduction

Independent pricing algorithms can coordinate through market feedback even without communication. Existing results emphasize reward signal quality, parallel experimentation, and the scoring rule faced by each learner (Hansen et al., 2021; Calvano et al., 2020). A different friction appears when the market itself remembers actions. A price cut can pay poorly today, yet become profitable only after enough consecutive cuts erase a router penalty. No single experiment reveals the long-run return.

AI inference routing provides a concrete instance. Providers sell execution of a common open-weight model, while a router combines posted price with hidden eligibility and service scores. On 22,330 public-menu snapshots for seven models, benchmark-reset counterfactuals imply 1.40–10.98 percentage points of price-induced active-provider share; these are rule quantities, not realized flow. A prospective owned-choice panel now estimates the latent score but has not crossed its support gate. We ask what learning problem arises when such a score itself remembers price actions.

This distinction matters for both algorithmic-collusion and mechanism-design claims. A high learned price can arise because high pricing is an equilibrium, because experiments are statistically coupled, or because a learner never discovers a profitable delayed path. These mechanisms demand different interventions. More exploration addresses the last mechanism only at an exponential cost; changing the action interface can solve it but may induce overcommitment.

Our contributions are fourfold.

1. We derive an exact *rational* boundary M^* for persistent cutting under a finite-memory penalty. It separates environments where the long-run low price is optimal from those where the high price is optimal.
2. We derive a distinct *learning* boundary. After any stopping time σ, an adaptive policy whose cut probability is at most q has conditional probability at most Tq^M of producing a fresh length-M path within T steps. Combining this bound with ordinary reward estimation separates late temporal credit from reward SNR.
3. We show that a length-$(M+1)$ commitment option preserves the optimal value and feasible primitive paths while replacing exponential path-proposal time by a linear term. This is an interface result, not a change in provider payoffs.
4. We test the mechanism in a frozen, calibrated finite MDP. The option closes the local gap at intermediate memory and becomes harmful beyond the rational boundary. A frozen trace replay separates early discovery from late credit starvation. State observability, cross-market severity, and a conventional multi-step return each fail prespecified tests.

2 Routing game and economic boundary

One market is a model–workload pair. Provider i posts price p_i, has marginal cost c_i, and receives first-route share

$$s_i(p, \alpha) = \frac{p_i^{-a} e^{\alpha_i}}{\sum_j p_j^{-a} e^{\alpha_j}}, \qquad a > 0. \tag{1}$$

The latent score α_i is reduced-form: quality, health, capacity, eligibility, and router preference may all enter. Defining $p_i^{\text{eff}} = p_i e^{-\alpha_i/a}$ makes equation (1) exactly an inverse-power rule in effective prices. Relative score obeys

$$\alpha_i - \alpha_j = \log(s_i/s_j) + a\log(p_i/p_j). \tag{2}$$

With fixed score and demand, profit is $s_i(p,\alpha)(p_i - c_i)$ and $\partial \log s_i/\partial \log p_i = -a(1-s_i)$. If instead $\alpha_i = h_i(\log p_i)$, the negative elasticity becomes $(1-s_i)(a-h_i')$: score feedback can attenuate or amplify the value of a cut. At a symmetric interior equilibrium with equal fixed scores,

$$p^* = c\frac{a(n-1)}{a(n-1)-n}, \tag{3}$$

which exists only if $a(n-1) > n$. The deployed value $a = 2$ is the duopoly knife edge. This static fact motivates the environment but is not our main learning result.

Now fix rivals and restrict one provider to a high quote H and low quote L. A low quote receives score $\alpha = \log\theta < 0$ until the previous M quotes are also low, after which its score returns to zero. This is equivalent to a temporary increase in effective price from L to $L\theta^{-1/a}$ even though the displayed quote is unchanged. Write u_H for profit at H, $u_{\theta L}$ for penalized profit at L, and u_L for unpenalized profit at L. The economically interesting ordering is

$$u_L > u_H > u_{\theta L}. \tag{4}$$

Thus cutting loses while the flag is active but wins after it clears.

Theorem 1 (Rational memory boundary)**.** *From an all-high history, an infinite-horizon optimal policy either stays high forever or cuts immediately and stays low. For discount $\gamma \in (0,1)$, cutting is strictly optimal iff*

$$\gamma^M > \frac{u_H - u_{\theta L}}{u_L - u_{\theta L}}. \tag{5}$$

The continuous boundary is $M^* = \log[(u_H - u_{\theta L})/(u_L - u_{\theta L})]/\log\gamma$.

In the frozen calibrated profile, $(u_H, u_{\theta L}, u_L) = (0.1067535, 0.0351280, 0.1501829)$ and $\gamma = 0.95$, giving $M^* = 9.2402$. Exact dynamic programming cuts for integer $M \le 9$ and stays high for $M \ge 10$. This boundary is derived before examining learning outcomes.

3 Critical memory for finite-time learning

Let $A_t \in \{H, L\}$. Fix any stopping time σ representing a stage at which downstream reward or value information has become usable, and define $\tau_M(\sigma)$ as the first $t \ge \sigma + M$ for which the M actions after σ ending at t are all low. This definition is intentionally conditional: an early visit can reveal the terminal reward without supplying a later one-step learner enough fresh support to propagate that value upstream. We allow full history dependence and require only

$$\mathbb{P}(A_t = L \mid \mathcal{F}_{t-1},\, t > \sigma,\, \tau_M(\sigma) \ge t) \le q. \tag{6}$$

Theorem 2 (Exponential fresh-path barrier)**.** *Under equation* (6)*, for every horizon T and almost every realized $\mathcal{F}_\sigma$,*

$$\mathbb{P}(\tau_M(\sigma) \le \sigma + T \mid \mathcal{F}_\sigma) \le (T - M + 1)_+ q^M \le Tq^M. \tag{7}$$

Consequently, attaining conditional fresh-path probability at least $1-\delta$ requires

$$T \ge (1-\delta)q^{-M}. \tag{8}$$

If post-σ actions are iid Bernoulli(q), the exact expected additional time is

$$\mathbb{E}[\tau_M(\sigma) - \sigma \mid \mathcal{F}_\sigma] = \frac{1-q^M}{(1-q)q^M}. \tag{9}$$

Define the finite-horizon critical memory

$$M_c(T, q, \delta) = \left\lfloor \frac{\log[T/(1-\delta)]}{\log(1/q)} \right\rfloor. \tag{10}$$

For $M \ge M_c + 1$, the target fresh-path probability is ruled out after any chosen readiness time σ. Thus memory is a computational control parameter for learners that need fresh downstream support: increasing T by a factor of ten buys only $1/\log_{10}(1/q)$ additional memory periods.

This clarifies the relationship to signal-to-noise results. Suppose reward differences are sub-Gaussian with gap Δ and proxy variance σ^2. Then $O(\mathrm{SNR}^{-1}\log(1/\delta))$ samples, where $\mathrm{SNR} = \Delta^2/\sigma^2$, suffice to identify the better downstream action. If upstream credit requires a fresh path after that readiness stage, equation (8) still applies.

Corollary 3 (Temporal–statistical crossover)**.** *For a fixed-confidence ordered-credit learner, total difficulty contains a fresh-support term q^{-M} and a statistical term $\mathrm{SNR}^{-1}\log(1/\delta)$. Temporal discovery dominates once*

$$q^{-M} \gg \mathrm{SNR}^{-1}\log(1/\delta). \tag{11}$$

Higher reward SNR cannot remove this conditional gap; it moves the readiness time σ but does not change post-σ path probability.

The result complements Hansen–Misra–Pai: independent experiments may couple through the final pricing score even with no explicit memory (Hansen et al., 2021). Our theorem does not contradict asymptotic learnability. It supplies a finite-time rate: once market memory requires a run rather than an isolated experiment, the time needed to re-expose the relevant score after downstream learning can be exponential in M. This conditional statement is not, by itself, an explanation of the Q-learning result; the trace audit below tests whether late support is actually absent.

4 A path-equivalent intervention

Augment the primitive MDP with a `commit-low` option that executes $M+1$ existing low actions. Both primitive actions remain available.

Theorem 4 (Value preservation and linear discovery)**.** *The augmented semi-Markov MDP has exactly the same optimal value as the primitive MDP at every state. It creates no new primitive price path. If the option is selected independently with probability q_o at each decision epoch while non-option decisions consume one transition, its completion time has expectation*

$$M + q_o^{-1}. \tag{12}$$

The option changes representation, not incentives. Any augmented policy can be unrolled into the same sequence of primitive actions and discounted rewards. But the probability of proposing the full low path is now q_o rather than q^M. This is the exponential temporal-abstraction gap shown in figure 1A.

5 Experimental design

The experiment is a controlled finite-MDP mechanism test, not a reduced-form test on provider prices. Its high and low quotes and router penalty are frozen from an audited market profile. Exact value iteration supplies the target action. Tabular Q-learning uses 300,000 primitive transitions, discount 0.95, and 20 prespecified seeds. The option learner receives the same primitive transition budget; an option consumes its full duration. A 10,000-transition frozen-policy evaluation reports first-action agreement and normalized regret. Paired seed bootstrap intervals describe simulation variation only.

We use a falsification ladder rather than interpreting one favorable run. E-SIM5 gives the primitive learner the complete Markov history. E-SIM6 adds only the path-equivalent option and sweeps $M \in \{1, 3, 5, 7, 9, 12\}$. E-SIM7 transports the intervention to four frozen price books. E-SIM8 varies learning rate and exploration decay in a 3×3 grid. E-SIM9 replaces one-step Q with an eight-step TD target while retaining the primitive action set. All gates, seeds, and result hashes were frozen before interpretation.

After freezing E-SIM6, we added a descriptive trace audit rather than modifying its confirmatory gate. It replays the exact 20 primitive seeds, asserts every terminal Q-table against the frozen artifact to tolerance 10^{-12}, records trailing-low-state visits after steps 100,000 and 200,000, and reconstructs the deterministic empirical transition table from visited pairs. Batch Bellman sweeps then test whether coverage is sufficient when update order is repaired. The checkpoints are diagnostics, not preregistered readiness estimates.

Displayed-price and latent-score calibration. The public calibration layer evaluates 22,330 snapshots for seven models at a fixed request shape and point exponent $a = 1.6483$. Resetting active quotes to the model-author benchmark removes median active shadow share ranging from 1.40 points for Kimi K2.6 to 10.98 points for GLM-5.2. The latter has a 27.3% median equivalent discount and transfers 5.40 points from anchor-priced providers under the rule. These exact menu counterfactuals calibrate the possible payoff scale; they do not measure realized provider traffic.

The complementary GLM-5.2 panel uses fresh owned choices to estimate equation (2) at the frozen exponent, plus a randomized explicit price-sort arm. It starts prospectively at 21 July 22:00 UTC, excludes four earlier observed choices, and reports nothing until at least 40 choices, 20 blocks, and three selected providers. Publication requires 800 choices, 100 blocks, seven days, and 90% menu coverage. At this freeze it is accruing with zero eligible choices. Even after the gate, a level score wedge calibrates $(u_H, u_{\theta L}, u_L)$ but does not identify the memory length M or validate the live temporal mechanism.

The live empirical bridge is deliberately weaker. In the latest hosted panel, the primary same-regime covariance statistic exceeds its randomization null by 0.0256 with one-sided $p = 0.0500$, but one provider-pair accounts for 87.3% of eligible events, the identity-shuffle diagnostic has $p = 0.992$, and downstream family gates are underpowered. This supports studying signal-coupling mechanisms; it does not validate them as provider conduct.

6 Results

Critical memory is visible in learning. At $M \in \{1, 3, 5\}$, primitive and option learners agree with exact control in 20/20 seeds. At $M = 7$, where exact optimization still cuts, primitive Q agrees in 1/20 seeds while option Q agrees in 19/20; 18/20 option seeds also satisfy the joint low-regret gate. At $M = 9$, the comparison is 0/20 versus 18/20. The transition is therefore near, but below, the independently derived rational boundary.

At $M = 7$, mean normalized regret is 0.07175 for primitive Q and 0.00750 for option Q. The paired difference is -0.06425 with 95% seed-bootstrap interval $[-0.07553, -0.04930]$. The sign remains negative in every one of nine learning-rate/exploration-decay cells; the strict composite severity gate passes in seven.

Early discovery is not the explanation. The trace audit reproduces all 20 frozen Q-tables exactly. Every primitive learner reaches the all-low state early: first-hit times range from step 7 to 901. Every seed also observes every one of the 256 state–action pairs at least nine times. Nevertheless, 19 finish with the wrong high initial action. Among those failures, none visits a state with five or more trailing low actions after step 100,000, and none returns

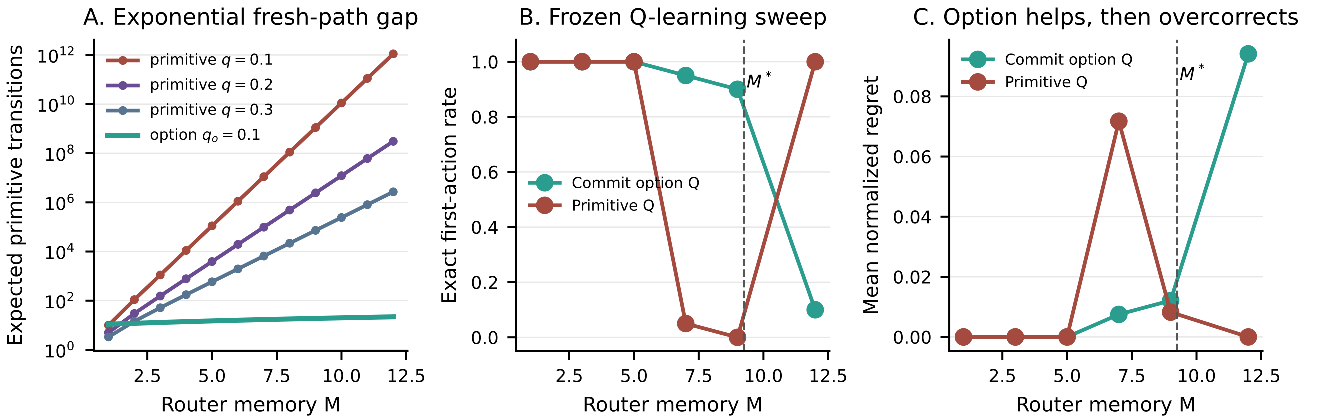

Figure 1: Theory and frozen experiment. A: after any readiness checkpoint, iid primitive exploration takes exponential additional time to produce a fresh length-M cut path; a commitment option is linear in M. B: exact first-action agreement over 20 seeds. C: mean normalized regret. The dashed rational boundary $M^* = 9.2402$ is computed from payoffs, not fitted to learner outcomes.

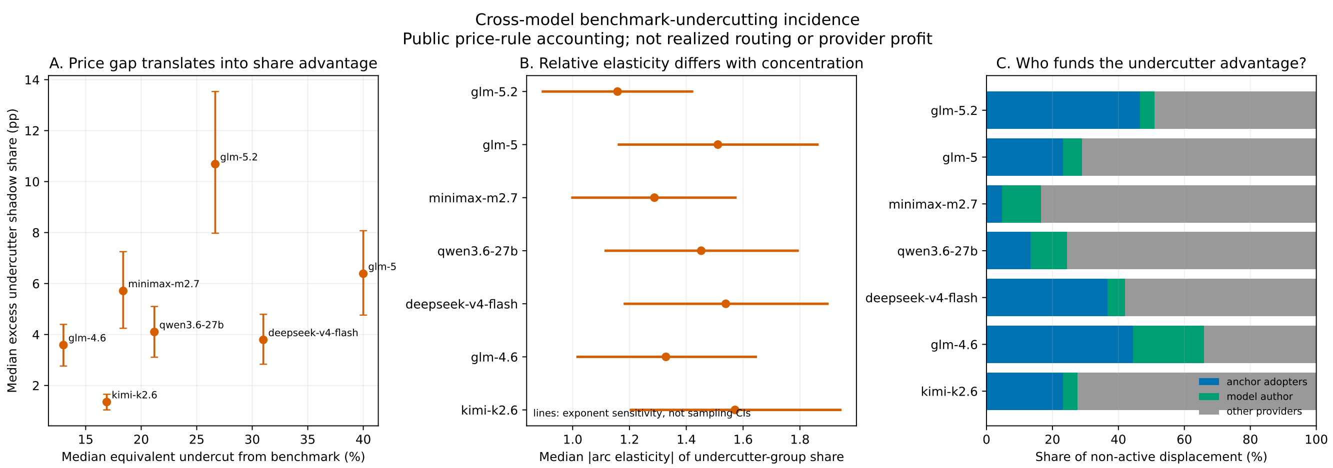

Figure 2: Public-price calibration across seven models. Benchmark resets identify deterministic share transfers under a declared inverse-power rule, not realized market share. The prospective owned-choice layer is required to estimate the price-equivalent latent score and remains support-gated.

to the all-low state. Thus neither first discovery nor aggregate support is the explanation. Ordered batch Bellman sweeps on each seed's same observed deterministic transitions recover the exact initial cut in 20/20 seeds. The observed mechanism is therefore *late credit starvation*: the data needed to solve the model were collected while exploration was broad, but online one-step updates did not revisit deep states after downstream values separated. This diagnostic motivates the stopping-time formulation of theorem 2; it does not estimate a unique readiness time.

The intervention is nonmonotone. At $M = 12$, exact optimization stays high. Primitive Q is correct in 20/20 seeds, whereas option Q is correct in only 2/20 and adds 0.0942 normalized regret. An interface that solves delayed credit can therefore reduce welfare when it creates commitment salience on the wrong side of the economic boundary. Mechanism evaluation must combine theorem 1 with theorem 2; the learning boundary alone is not a design objective.

The falsifications narrow the mechanism. Complete state observability does not repair primitive learning: it remains high in 19/20 seeds at $M = 7$. The severe-trap transport gate fails in two of four external price books, although all four option-effect signs align with their rational boundary. Finally, an ordinary eight-step TD target succeeds in 0/20 seeds and has regret difference $+0.0038$ relative to one-step Q, interval $[0, 0.0113]$. The supported result is option-specific temporal abstraction in the controlled MDP, not a claim that any longer return solves delayed credit.

7 Implications for pricing mechanisms

The two boundaries separate three regimes.

1. If $M > M^*$, a persistent cut destroys provider value. High pricing is rational; exploration policy is secondary.
2. If $M \le M^*$ but $M > M_c(T, q, \delta)$, cutting is rational but is unlikely to receive a fresh length-M support path within the late-learning horizon. A platform can sustain high learned prices without communication or supra-rational reward coupling.
3. If $M \le \min\{M^*, M_c\}$, ordinary exploration can expose the low path; reward SNR and cross-agent signal coupling again become first-order.

This yields a design warning. Router memory can look anti-gaming because it disciplines short-lived cuts, yet the same rule can suppress efficient persistent cuts by altering learnability. Conversely, exposing a commitment action can restore price competition below M^* but cause excessive cutting above it. A safer router would expose the state and a cancellable multi-period quote schedule only when a deviation audit predicts that persistent cutting raises provider value and buyer surplus. The theorem supplies a testable screen, not a universally optimal policy.

The target score must also match the economic objective. A user-cost router scores payment, latency, and failure; a welfare router substitutes resource cost for payment and values delivered quality; a revenue router may favor high spend under ad-valorem fees; and a quality-priority router maximizes verified fidelity subject to budget. These objectives generate different payoff triples and hence different M^*. The commitment option should be evaluated on a Pareto frontier over user cost, welfare bounds, router revenue, quality, completion, and provider exploitability rather than on price alone.

Provider technology shifts the same boundary. An owned or reserved-capacity provider pays high fixed cost but may rationally cut to maintain utilization; a spot-dependent provider has less sunk exposure but state-dependent scarcity cost; an anchor adopter may not update displayed price at all. A router that hides score memory can therefore trap low-short-run-cost providers while doing little to passive anchors. Practical implementation should expose the score state, offer cancellable multi-period quote commitments, use execution-contingent capacity rather than fundraising or provider identity as a proxy, and charge fees separately from selected spend.

8 Related work

Algorithmic-pricing research establishes that independent learners may support high prices and that results depend sharply on algorithms and timing (Calvano et al., 2020; Asker et al., 2022; Brown and MacKay, 2023). Hansen–Misra–Pai show how independent bandit experiments become strategically coupled through the aggregate price used to score arms (Hansen et al., 2021). Johnson–Rhodes–Wildenbeest study platform steering when sellers use pricing algorithms (Johnson et al., 2023). Our contribution is orthogonal: a router's own-history rule creates a finite-time fresh-support barrier even when reward differences are otherwise easy to estimate.

Options and semi-Markov value preservation are classical (Sutton et al., 1999); delayed reward and sequence compression have recently been studied directly (Han et al., 2022; Ramesh et al., 2024). We add an economic source of the delay, an explicit critical-memory rate, a rational boundary with the opposite comparative-static force, and controlled evidence that the representation intervention helps only between those boundaries.

9 Limitations and claim boundary

The finite-time theorem assumes a post-stopping-time cut-probability cap and applies only to learners that require fresh downstream support. It does not characterize algorithms that plan over a known model, retain replay sufficient for exact offline backups, or deliberately schedule low runs. The diagnostic checkpoint is post hoc and does not identify when a Q-value becomes usable. The calibrated experiment fixes rivals, uses two prices, and studies tabular Q. Twenty seeds identify Monte Carlo variability, not uncertainty in live-market parameters. Cross-market severe-trap transport fails, and no paid probe randomizes router memory. The empirical covariance screen is concentrated and family-incomplete. We therefore claim a theorem, a controlled mechanism, and a reproducible local counterfactual—not provider intent, tacit collusion, communication, or live-router causality.

10 Conclusion

Market memory creates two distinct thresholds. The rational threshold says whether persistent cutting is valuable; the finite-time threshold says whether a bounded ordered-credit learner can obtain fresh support after its downstream values become usable. Between them, high learned prices can persist despite a profitable low-price path, with late path time exponential in memory and only logarithmic improvement from longer training. Complete state–action coverage is insufficient for the online learner, while ordered batch backups on those same transitions recover the optimum. A path-equivalent option also collapses that gap but overcorrects when cutting is no longer rational. Learning-aware marketplace design must therefore optimize both incentives and representations, then audit deviations on either side of the boundary.

A Proofs

Proof of theorem 1. Once the history is all-low, L strictly dominates H by equation (4); choosing H also restarts the penalty. Before that state, any policy that eventually cuts is weakly improved to k high actions followed by low forever. Its value is

$$V_k = \frac{u_H(1-\gamma^k)}{1-\gamma} + \gamma^k V_{\text{cut}},$$
$$V_{\text{cut}} = \frac{(1-\gamma^M)u_{\theta L} + \gamma^M u_L}{1-\gamma}.$$

Because $V_k - u_H/(1-\gamma) = \gamma^k[V_{\text{cut}} - u_H/(1-\gamma)]$, either $k = 0$ or never cutting is optimal. Comparing these alternatives yields equation (5). □

Proof of theorem 2. Condition on $\mathcal{F}_\sigma$. For each candidate ending time $t > \sigma$, iterated conditioning and equation (6) bound the probability of the required post-σ all-low length-M window by q^M. A union bound over $(T - M + 1)_+$ windows proves equation (7); inversion of its looser Tq^M form gives equation (8). Under iid Bernoulli exploration, let E_k be the expected remaining time after a current run of k lows. The recursions $E_M = 0$ and $E_k = 1 + qE_{k+1} + (1-q)E_0$ solve to $E_0 = (1-q^M)/[(1-q)q^M]$. □

Proof of theorem 4. The augmented action set contains every primitive action, so its optimal value is weakly larger. Replace each option in any augmented policy by its defining $M+1$ primitive low actions. The unrolled policy produces identical states, durations, and discounted rewards, so its value is identical and the augmented optimum cannot be larger. Before the first option, the number of one-step decisions is geometric with mean $(1-q_o)/q_o$; adding the final $M+1$ transitions gives $M + q_o^{-1}$. □

B Reproducibility

The theory functions and unit tests are in `src/orcap/market_env/critical_memory.py` and `tests/market_env/test_critical_memory.py`. The figure is generated directly from the frozen E-SIM6 parquet. Clean bundles are E-SIM5 `3e5a55405e`, E-SIM6 `93265b9b03`, E-SIM7 `bd74ab2eb7`, E-SIM8 `4d84b9b3a2`, and E-SIM9 `179eca0f9d`. Manifests pin source commits, seeds, result hashes, and input-artifact hashes. Exact Bellman residuals, path equivalence, paper numbers, and the new finite-time identities are enforced by tests. The post-hoc trace artifact `credit-propagation-diagnostic.json` is generated by the adjacent replay script. It is admitted only after all 20 Q-tables match the frozen E-SIM6 artifact; separate tests reconcile seed-level actions, early hits, and late depth counts. The public price-transfer and prospective latent-score claims are frozen in `../price-score-evidence-2026-07-21.json`; the latter is explicitly an accruing design with no numerical score result. Its remote monitor conditions on exact owned-request menus and never pools H81 or H95 outcomes.

References

John Asker, Chaim Fershtman, and Ariel Pakes. Artificial intelligence, algorithm design, and pricing. *AEA Papers and Proceedings*, 112:452–456, 2022. doi: 10.1257/pandp.20221058.

Zach Y. Brown and Alexander MacKay. Competition in pricing algorithms. *American Economic Journal: Microeconomics*, 15 (2):109–156, 2023. doi: 10.1257/mic.20210158.

Emilio Calvano, Giacomo Calzolari, Vincenzo Denicolò, and Sergio Pastorello. Artificial intelligence, algorithmic pricing, and collusion. *American Economic Review*, 110(10):3267–3297, 2020. doi: 10.1257/aer.20190623.

Beining Han, Zhizhou Ren, Zuofan Wu, Yuan Zhou, and Jian Peng. Off-policy reinforcement learning with delayed rewards. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 8280–8303, 2022.

Karsten T. Hansen, Kanishka Misra, and Mallesh M. Pai. Frontiers: Algorithmic collusion: Supra-competitive prices via independent algorithms. *Marketing Science*, 40(1):1–12, 2021. doi: 10.1287/mksc.2020.1276.

Justin P. Johnson, Andrew Rhodes, and Matthijs R. Wildenbeest. Platform design when sellers use pricing algorithms. *Econometrica*, 91(5):1841–1879, 2023. doi: 10.3982/ECTA19978.

Aditya Ramesh, Kenny John Young, Louis Kirsch, and Jürgen Schmidhuber. Sequence compression speeds up credit assignment in reinforcement learning. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 42104–42126, 2024.

Richard S. Sutton, Doina Precup, and Satinder Singh. Between MDPs and semi-MDPs: A framework for temporal abstraction in reinforcement learning. *Artificial Intelligence*, 112(1–2):181–211, 1999. doi: 10.1016/S0004-3702(99)00052-1.